

Contents

1	Introduction	4
1.1	System Overview	4
1.2	Purpose	4
1.3	Scope	4
1.4	Definitions	4
1.5	Constraints	5
2	System Architecture	6
2.1	System Overview	6
2.2	Three-Tier Architecture	6
2.3	Preprocessing Pipeline	7
2.4	Enrollment Process	7
2.5	Verification Process	9
2.6	Processing Pipeline	9
2.7	Deepfake Generation for Testing	10
3	Literature Survey	11
3.1	Speaker Verification	11
3.2	Audio Deepfake Detection	11
3.3	Model Comparison	11
3.4	Model Specifications	11
3.4.1	Speaker Recognition: ReDimNet-B5	11
3.4.2	Deepfake Detection: AASIST (Under Evaluation)	12
4	Design	13
4.1	Data Design	13
4.1.1	Database Description	13
4.1.2	Global Data Structures	13
4.2	Structural Design	14
4.2.1	Class Diagram	14
4.3	Interactions Design	15
4.3.1	Use Case Diagram	15
4.3.2	Sequence Diagram: Enrollment	15
4.3.3	Sequence Diagram: Verification	15
4.3.4	Sequence Diagram: TCAV Explanation	16
4.3.5	Activity Diagram: Verification	16
4.4	Software Architecture Pattern	18
4.4.1	MVC Structure	18
4.5	Testing Platform	18
5	User Interface	19
5.1	Enrollment Screen	19
5.2	Processing Screen	19
5.3	Deepfake Detection Result Screen	21
5.4	Verification Result Screen	21
5.5	TCAV Explanation Screen	23
5.6	Deepfake Generator (Testing)	23

6	Appendix-A: TCAV Explainability	25
6.1	Acoustic Concept Taxonomy	25
6.2	Layer-wise Analysis Results	25
6.3	TCAV Integration	25
7	Appendix-B: Team	25
8	Appendix-C: Schedule	25
	References	27

List of Figures

1	High-level system architecture showing main components and data flow	6
2	Three-tier architecture of the BioVoice system	6
3	Audio preprocessing pipeline stages	7
4	Enrollment process flow diagram	8
5	Verification process flow with parallel speaker verification and deepfake detection .	9
6	Complete processing pipeline with intermediate representations	10
7	Data schema for profile and configuration storage	13
8	Class diagram showing main system components	14
9	Use case diagram showing actor interactions	15
10	Sequence diagram for enrollment process	15
11	Sequence diagram for verification process	16
12	Sequence diagram for TCAV explanation request	16
13	Activity diagram for verification process	17
14	Model-View-Controller architecture	18
15	Enrollment screen with voice recording interface	19
16	Audio processing screen showing pipeline progress	20
17	Deepfake detection result showing genuine audio analysis	21
18	Verification result screen showing successful authentication	22
19	TCAV explanation screen showing voice concept contributions	23
20	Deepfake generator screen for testing and validation	24

List of Tables

1	Technical terminology and definitions	5
2	Preprocessing pipeline parameters	7
3	Processing pipeline stage specifications	10
4	Speaker recognition model comparison on VoxCeleb benchmark	11
5	ReDimNet-B5 specifications	12
6	Global data structures	13
7	Testing requirements and acceptance criteria	18
8	Acoustic concept attributes	25
9	Layer-wise TCAV analysis results (STAGE4 optimal)	25
10	Project team members	25
11	Project schedule	26

1. Introduction

1.1 System Overview

BioVoice is a voice biometric authentication system designed for speaker identity verification and synthetic audio detection.

Authentication systems are increasingly adopting biological characteristics for identity verification. Biometric features such as fingerprints, facial geometry, and voice patterns offer advantages over traditional knowledge-based authentication methods (passwords, PINs) which can be compromised through theft or social engineering. Voice biometrics presents a promising modality for remote authentication, though the uniqueness and reliability of voice as a biometric identifier requires rigorous validation. This project investigates and implements voice-based authentication while addressing the emerging challenge of AI-generated synthetic audio.

With the rise of AI-generated voices and deepfakes, we also consider protection against synthetic audio. So not only does BioVoice aim at recognizing the speaker, it also aims at recognizing whether it's really a human or a machine trying to fool it.

The BioVoice system implements the following core functions:

- **User Enrollment** — The system captures a voice sample and extracts a speaker embedding to create a stored voice profile.
- **Identity Verification** — The system compares a test voice sample against the stored profile using cosine similarity to authenticate the claimed identity.
- **Deepfake Detection** — If someone tries to use AI-generated audio to impersonate a user, the system flags it.
- **Explainable Decisions** — Using TCAV (Testing with Concept Activation Vectors), the system provides interpretable explanations for verification decisions by identifying acoustic patterns that contributed to the outcome.

1.2 Purpose

This Software Design Document (SDD) specifies the technical design of the BioVoice voice biometric authentication system. The document describes the system architecture, component design, data structures, processing algorithms, and user interfaces. It serves as a technical reference for system implementation, review, and maintenance.

1.3 Scope

BioVoice is an application designed for identity verification using voice biometrics combined with deepfake detection.

System Capabilities:

- Desktop application for voice-based identity verification
- One-to-one speaker verification (verifying a claimed identity against a stored profile)
- Deepfake and synthetic audio detection
- Explainable AI module providing interpretable verification decisions

1.4 Definitions

Table 1 provides definitions for technical terms used throughout this document.

Table 1: Technical terminology and definitions

Term	Definition
Authentication	Proving you are who you claim to be
Biometrics	Using body features (face, fingerprint, voice) to identify someone
Enrollment	One-time setup where you record your voice so the system remembers you
Verification	Checking if a new voice sample matches a stored profile
Sample Rate	How many times per second we measure sound (we use 16,000/sec)
Spectrogram	A visual picture of sound showing pitch over time
Mel-Spectrogram	A spectrogram tuned to how human ears hear sound
Neural Network	An AI system that learns patterns from examples
Model	A trained AI ready to make predictions
ReDimNet-B5	Our speaker recognition AI — turns voice into a 192-number fingerprint
AASIST	Our deepfake detector AI — catches fake/synthetic audio
TCAV	Testing with Concept Activation Vectors — an explainability method that identifies structural patterns in mel-spectrograms that are visually interpretable by humans, enabling understanding of <i>why</i> the model made a specific decision.
Embedding	A 192-number fingerprint representing your unique voice
Cosine Similarity	A mathematical measure of similarity between two embedding vectors. Computed as the cosine of the angle between them: $\cos(\theta) = \frac{\mathbf{A} \cdot \mathbf{B}}{\ \mathbf{A}\ \ \mathbf{B}\ }$. Values range from 0 (completely different) to 1 (identical). Used to compare a test embedding against the stored template.
Threshold	The cutoff score for the speaker verification decision. If cosine similarity ≥ 0.75 , the speaker is verified; otherwise, access is denied.
Deepfake	AI-generated fake audio pretending to be a real person
Spoofing	Trying to fool the system with fake audio
EER	Error rate metric — ours is $<1\%$ (wrong less than 1 in 100)
Latency	Wait time from speaking to getting a result (<2 seconds)
API	A way for software to talk to external services

1.5 Constraints

Audio requirements: We work with 16 kHz audio (the standard for speech processing). Recordings should be between 1 and 10 seconds — short enough to be convenient, long enough to capture your voice’s unique characteristics.

Accuracy targets: We’re aiming for less than 1% error rate on speaker verification and over 95% accuracy on deepfake detection. These are ambitious but achievable goals based on our model selection.

Speed: The entire process — from recording to decision — should take less than 2 seconds.

Environment: This is a desktop application. We assume a reasonably quiet environment and a decent microphone, though the system is designed to be robust to everyday background noise.

2. System Architecture

This section describes the overall architecture of the BioVoice system, including component organization, data flow, and processing pipelines.

2.1 System Overview

Figure 1 presents the high-level architecture of BioVoice, showing the main components and data flow from audio input to verification decision.

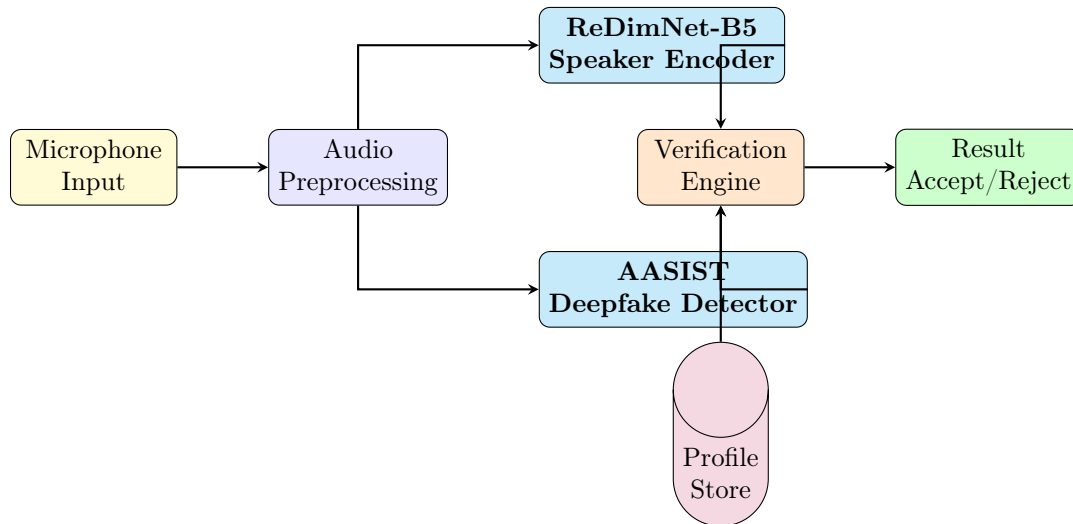


Figure 1: High-level system architecture showing main components and data flow

The system operates as follows: Audio input is captured via microphone and processed through the preprocessing pipeline to generate mel-spectrogram features. These features are processed in parallel by ReDimNet-B5 (for speaker embedding extraction) and AASIST (for deepfake detection). The Verification Engine compares the extracted embedding against the stored profile template and combines both scores to produce the final accept/reject decision.

2.2 Three-Tier Architecture

The BioVoice system follows a three-tier architecture pattern to separate concerns and improve maintainability. Figure 2 illustrates this structure.

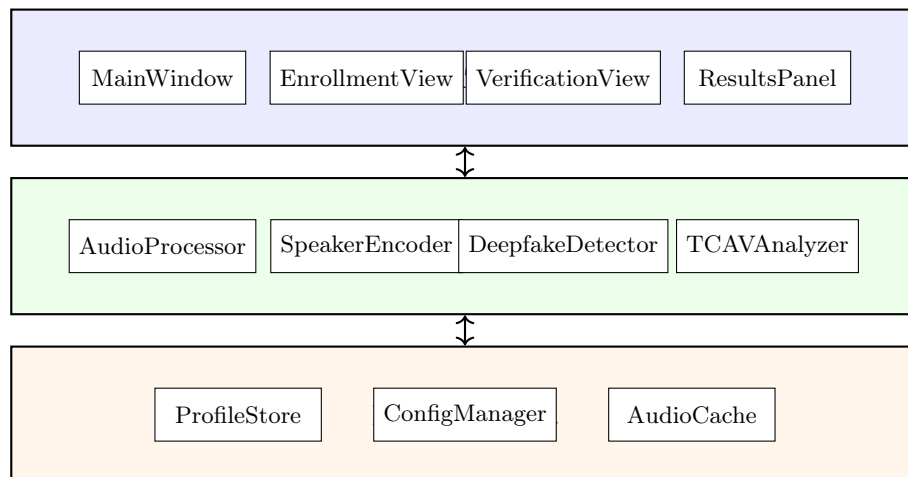


Figure 2: Three-tier architecture of the BioVoice system

Presentation Tier: Contains all user interface components including MainWindow (application shell), EnrollmentView (voice registration interface), VerificationView (authentication interface), and ResultsPanel (outcome display with similarity scores and explanations).

Logic Tier: Implements core processing functionality. AudioProcessor handles signal pre-processing (resampling, normalization, mel-spectrogram computation). SpeakerEncoder wraps the ReDimNet-B5 model for embedding extraction. DeepfakeDetector wraps the AASIST model for authenticity verification. TCAVAnalyzer provides explainability features by analyzing concept activations.

Data Tier: Manages persistent storage and caching. ProfileStore handles user voice profiles (embeddings and metadata). ConfigManager maintains system configuration parameters. AudioCache provides temporary storage for audio processing efficiency.

2.3 Preprocessing Pipeline

The preprocessing pipeline transforms raw audio input into mel-spectrogram features suitable for the neural network models. This transformation extracts frequency-domain representations that capture the acoustic characteristics of speech.

The pipeline operates through the following stages: First, the raw audio waveform is loaded from the microphone or file. If the sample rate differs from the target 16 kHz, the audio is resampled using linear interpolation. Stereo recordings are converted to mono by averaging the two channels. The amplitude is then normalized to the range $[-1, 1]$ to ensure consistent input levels regardless of recording volume. Finally, a mel-spectrogram is computed using 80 mel-frequency bands, a 512-sample FFT window with 160-sample hop length (10ms frame shift), covering the frequency range 20–8000 Hz with a Hann window function. The output is a tensor of shape $[80, T]$ where T represents the number of time frames, approximately 100 frames per second of audio.



Figure 3: Audio preprocessing pipeline stages

Table 2 lists the preprocessing parameters.

Table 2: Preprocessing pipeline parameters

Parameter	Value	Parameter	Value
Target sample rate	16,000 Hz	FFT size	512
Mel bands	80	Hop length	160
Frequency range	20–8000 Hz	Window	Hann

2.4 Enrollment Process

Enrollment is the process of creating a new speaker profile from a voice recording. During enrollment, the user provides a unique identifier and records a voice sample. The system validates that the identifier does not already exist in the database. The recorded audio is then processed through the preprocessing pipeline to generate a mel-spectrogram. ReDimNet-B5 extracts a 192-dimensional speaker embedding from the spectrogram, which is L2-normalized to unit length. The profile (embedding vector and metadata including user ID, creation timestamp, and sample count) is stored to disk for future verification. Figure 4 illustrates this process.

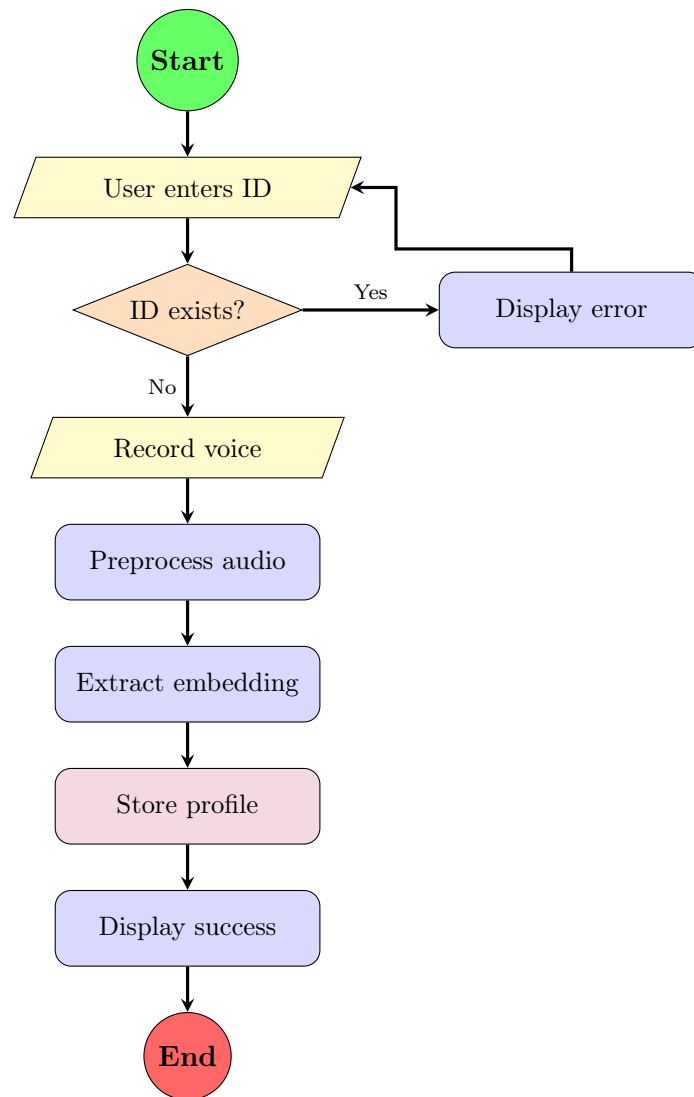


Figure 4: Enrollment process flow diagram

The enrollment process parameters are:

1. Number of samples required: 1
2. Sample duration: 1–10 seconds
3. Embedding dimension: 192
4. Storage per user: 768 bytes (embedding) + metadata
5. Total enrollment time: < 15 seconds

2.5 Verification Process

Verification is the process of authenticating a claimed identity by comparing a test voice sample against the stored profile. The process involves two parallel checks: speaker similarity verification using ReDimNet-B5, and deepfake detection using AASIST.

The verification flow operates as follows: The user selects their profile and provides a voice sample via microphone. The audio is preprocessed into a mel-spectrogram, which is then processed by both models in parallel. ReDimNet-B5 extracts a 192-dimensional embedding from the test audio. This embedding is L2-normalized and compared against the stored template using cosine similarity. Simultaneously, AASIST analyzes the mel-spectrogram to produce an authenticity score indicating whether the audio is genuine human speech or synthetic/manipulated audio. The final decision combines both scores: if the cosine similarity meets or exceeds 0.75 AND the AASIST score meets or exceeds 0.5, the user is authenticated. Otherwise, access is denied. Figure 5 illustrates this process.

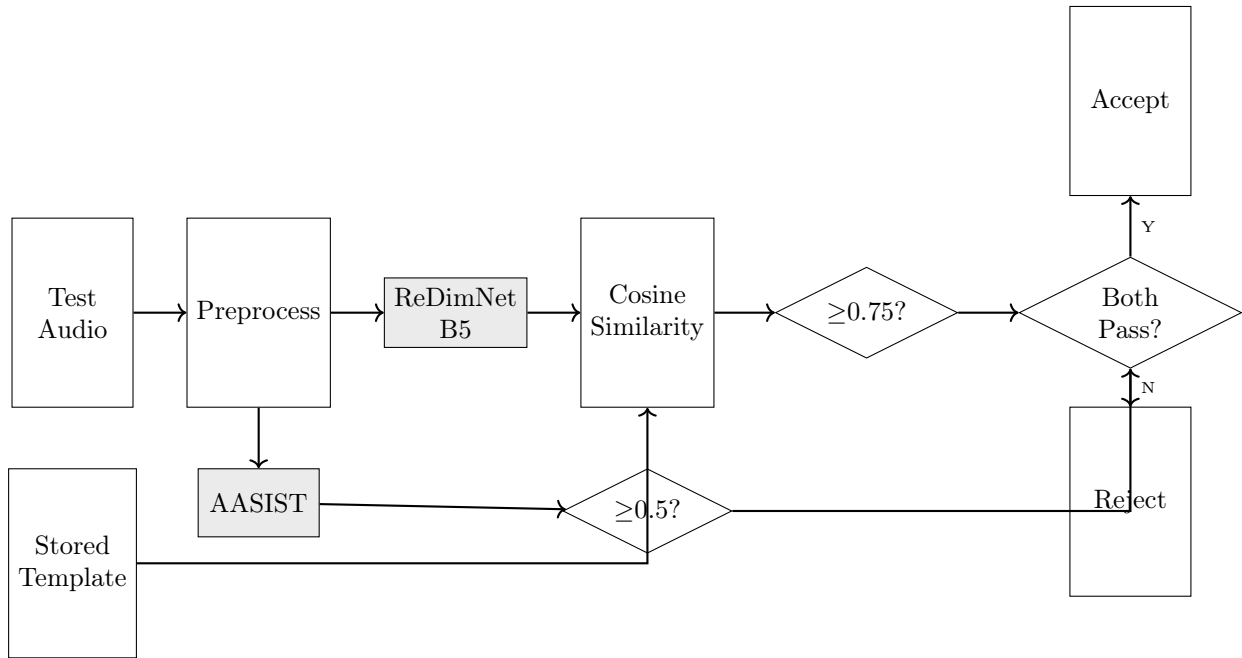


Figure 5: Verification process flow with parallel speaker verification and deepfake detection

The authentication logic is:

$$\text{ACCEPT} = (\text{similarity} \geq 0.75) \wedge (\text{deepfake_score} \geq 0.5)$$

2.6 Processing Pipeline

The complete processing pipeline integrates all stages from audio capture to final decision. Figure 6 shows the data flow through each component, including intermediate representations and latencies.

Audio is captured at 16 kHz sample rate and stored as PCM float32 samples. The preprocessing stage produces a clean, normalized audio signal, which is then transformed into a mel-spectrogram tensor of shape $[80, T]$. This representation feeds into both the speaker encoder (ReDimNet-B5, producing a 192-dimensional embedding) and the deepfake detector (AASIST, producing an authenticity score). The similarity computation compares the test embedding against the stored template using cosine similarity. Finally, the decision stage combines both scores to produce the accept/reject outcome.

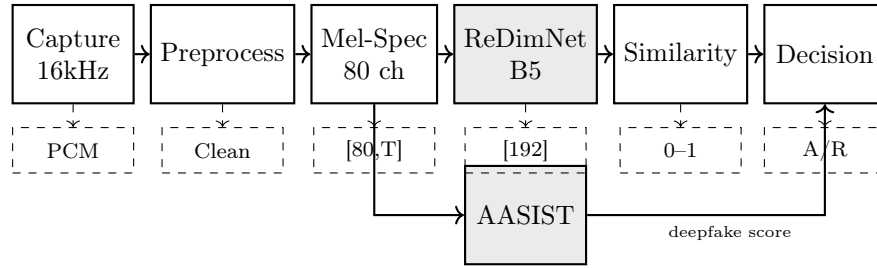


Figure 6: Complete processing pipeline with intermediate representations

Table 3 details the input, output, and latency of each pipeline stage.

Table 3: Processing pipeline stage specifications

Stage	Input	Output	Model	Latency
1. Capture	Microphone	PCM float32[]	—	1–10s
2. Preprocess	Raw audio	Clean audio	—	15ms
3. Features	Clean audio	Mel-spec [80, T]	—	10ms
4. Speaker Encoding	Mel-spec	Embedding [192]	ReDimNet-B5	25ms
5. Deepfake Detection	Mel-spec	Score [0, 1]	AASIST	30ms
6. Similarity	Embedding pair	Score [0, 1]	—	1ms
7. Decision	Both scores	Accept/Reject	—	1ms

2.7 Deepfake Generation for Testing

To validate the AASIST deepfake detection capability, the system includes a testing module that generates synthetic audio samples using external Text-to-Speech (TTS) and Voice Conversion (VC) APIs. This module is used exclusively for testing and benchmarking purposes.

The testing process operates as follows: A source audio sample of the target speaker is provided as input. The desired text to be synthesized is defined. An external TTS/VC API (such as ElevenLabs or Coqui TTS) generates a voice-cloned audio sample. The synthetic sample is saved and then fed to the AASIST detector to verify that it is correctly flagged as fake (score < 0.5). This capability ensures proper calibration of the deepfake detection module against state-of-the-art voice cloning systems.

3. Literature Survey

This section reviews prior work in speaker verification and audio deepfake detection, providing context for the model selection decisions in this project.

3.1 Speaker Verification

Speaker verification is the task of determining whether a speech sample belongs to a claimed speaker identity. Modern approaches use deep neural networks to extract speaker embeddings—compact vector representations that capture voice characteristics independent of linguistic content.

For this project, we evaluated three main approaches:

Wav2Vec 2.0: A self-supervised speech representation model that learns from unlabeled audio. While achieving good results, its large model size (95M parameters) makes it impractical for real-time desktop applications.

ECAPA-TDNN: An enhanced time-delay neural network architecture incorporating squeeze-excitation blocks and multi-scale features. It achieves 1.44% EER on VoxCeleb with only 6.2M parameters, offering a good balance of accuracy and efficiency.

ReDimNet: A family of speaker embedding models achieving state-of-the-art performance through efficient dimensionality reduction. ReDimNet-B5 was selected for this project based on its best-in-class EER of 0.79% combined with reasonable computational requirements (9.1M parameters).

3.2 Audio Deepfake Detection

Audio deepfake detection (anti-spoofing) aims to distinguish genuine human speech from synthetic or manipulated audio. This capability is essential as text-to-speech (TTS) and voice conversion technologies have become increasingly realistic.

For testing our detection capabilities, we generated synthetic audio samples using F5-TTS, a state-of-the-art text-to-speech model capable of producing highly realistic voice clones from short reference audio. These samples were used to evaluate detection performance.

For the detection module, we are evaluating AASIST (Audio Anti-Spoofing using Integrated Spectro-Temporal features), a model that uses graph neural networks to identify synthetic audio. AASIST analyzes spectro-temporal patterns to distinguish genuine speech from spoofed audio.

3.3 Model Comparison

Table 4 compares speaker verification models evaluated for this project. ReDimNet-B5 was selected based on its best-in-class EER of 0.79% combined with reasonable computational requirements.

Table 4: Speaker recognition model comparison on VoxCeleb benchmark

Model	EER	Params	GMACs	Notes
Wav2Vec 2.0	3.0%	95M	90	Too large
ECAPA-TDNN	1.44%	6.2M	2.5	Higher EER
ResNet-293	0.89%	25M	50	Too heavy
ReDimNet-B1	1.21%	1.9M	0.8	Higher EER
ReDimNet-B3	0.89%	4.2M	3.0	Close
ReDimNet-B5	0.79%	9.1M	10	Selected

3.4 Model Specifications

3.4.1 Speaker Recognition: ReDimNet-B5

ReDimNet-B5 [2] is a speaker embedding model that achieves state-of-the-art performance through efficient dimensionality reduction. The model takes 80-dimensional mel-spectrograms as input and

produces 192-dimensional L2-normalized embeddings. It was trained on VoxCeleb1 and VoxCeleb2 datasets and achieves an EER of 0.79% on VoxCeleb1-O. The architecture uses attentive statistics pooling to aggregate frame-level features into utterance-level embeddings.

Table 5: ReDimNet-B5 specifications

EER	0.79%	Parameters	9.1M
Input	80-dim mel-spec	Output	192-dim embedding
Architecture	Attentive Statistics Pooling	Training	VoxCeleb1+2

3.4.2 Deepfake Detection: AASIST (Under Evaluation)

AASIST (Audio Anti-Spoofing using Integrated Spectro-Temporal features) is a candidate model for detecting synthetic audio. The model processes mel-spectrograms using graph attention layers to capture spectro-temporal patterns that distinguish genuine speech from spoofed audio. AASIST outputs an authenticity score between 0 and 1, where higher scores indicate genuine audio. We are currently evaluating this model for integration into the BioVoice system.

4. Design

4.1 Data Design

4.1.1 Database Description

BioVoice uses file-based storage for user profiles and configuration. Figure 7 shows the data schema.

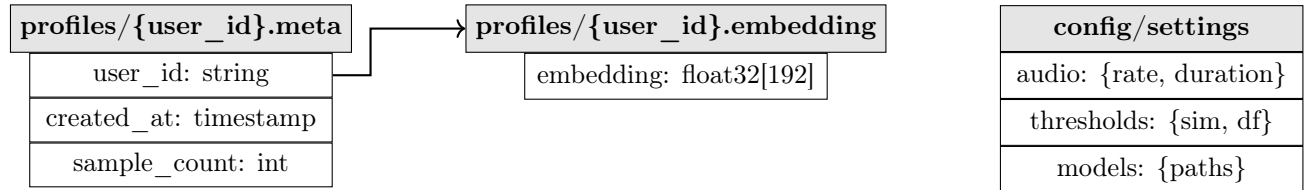


Figure 7: Data schema for profile and configuration storage

Storage Structure:

```

biovoice/
  profiles/
    {user_id}.meta      # Metadata
    {user_id}.embedding # 192-dim vector
  config/
    config.settings     # System configuration
  logs/
    verification.log    # Audit trail
  
```

4.1.2 Global Data Structures

Table 6 describes the main data structures used in the system.

Table 6: Global data structures

Structure	Type	Size	Description
AudioBuffer	float32[]	variable	Raw PCM samples at 16kHz
MelSpectrogram	float32[80][T]	$80 \times T$	Mel-frequency features
Embedding	float32[192]	768 B	L2-normalized speaker vector
VoiceProfile	struct	~1 KB	{user_id, embedding, metadata}
VerifyResult	struct	32 B	{auth, sim_score, df_score, time}

4.2 Structural Design

4.2.1 Class Diagram

Figure 8 shows the main classes and their relationships. The VerificationEngine serves as the central orchestrator.

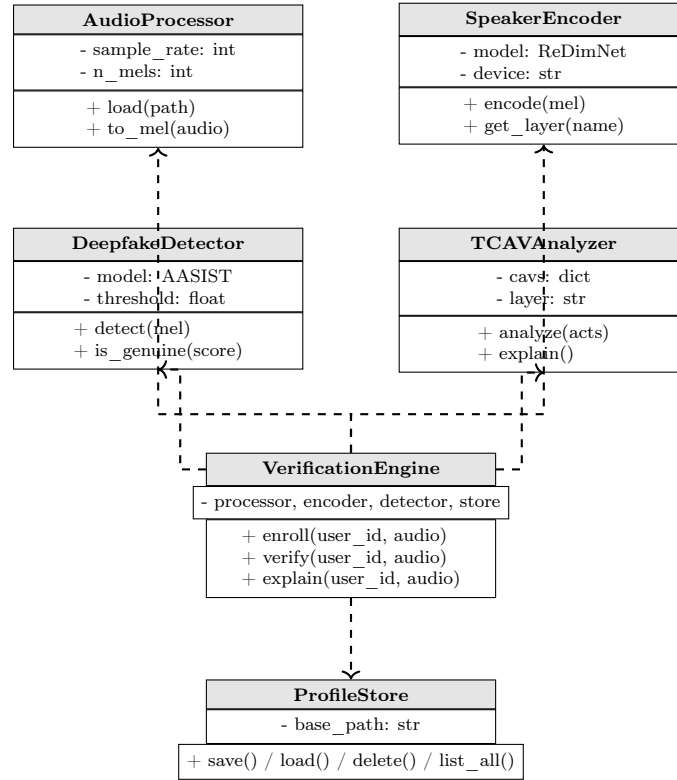


Figure 8: Class diagram showing main system components

4.3 Interactions Design

4.3.1 Use Case Diagram

Figure 9 shows the use case diagram for the BioVoice system.

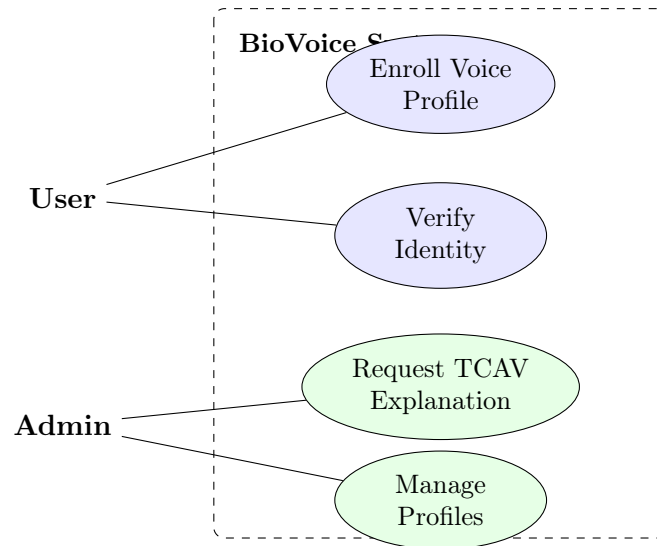


Figure 9: Use case diagram showing actor interactions

4.3.2 Sequence Diagram: Enrollment

Figure 10 shows the sequence diagram for the enrollment process.

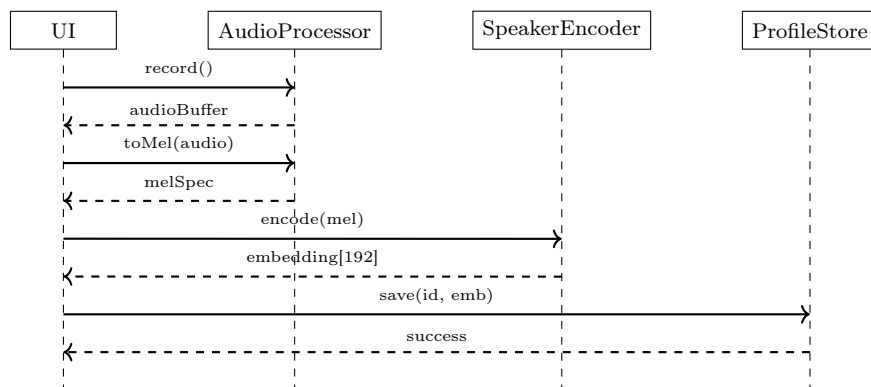


Figure 10: Sequence diagram for enrollment process

4.3.3 Sequence Diagram: Verification

Figure 11 shows the sequence diagram for the verification process.

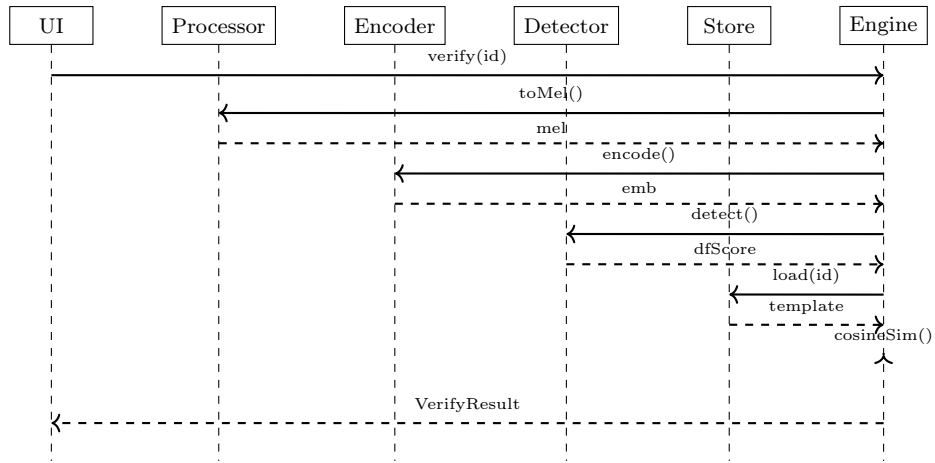


Figure 11: Sequence diagram for verification process

4.3.4 Sequence Diagram: TCAV Explanation

Figure 12 shows the sequence diagram for requesting a TCAV explanation.

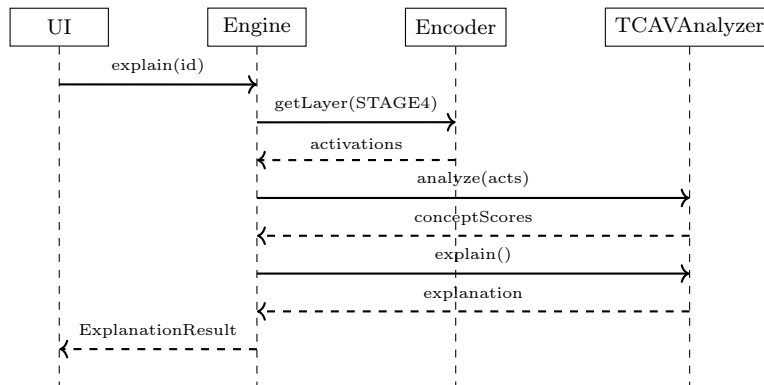


Figure 12: Sequence diagram for TCAV explanation request

4.3.5 Activity Diagram: Verification

Figure 13 shows the activity diagram for the verification process.

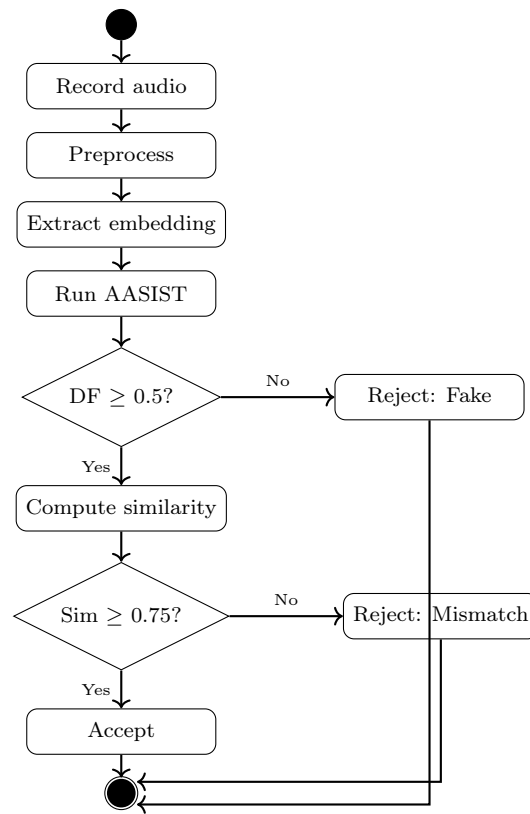


Figure 13: Activity diagram for verification process

4.4 Software Architecture Pattern

4.4.1 MVC Structure

The BioVoice system follows the Model-View-Controller (MVC) pattern. Figure 14 illustrates this architecture.

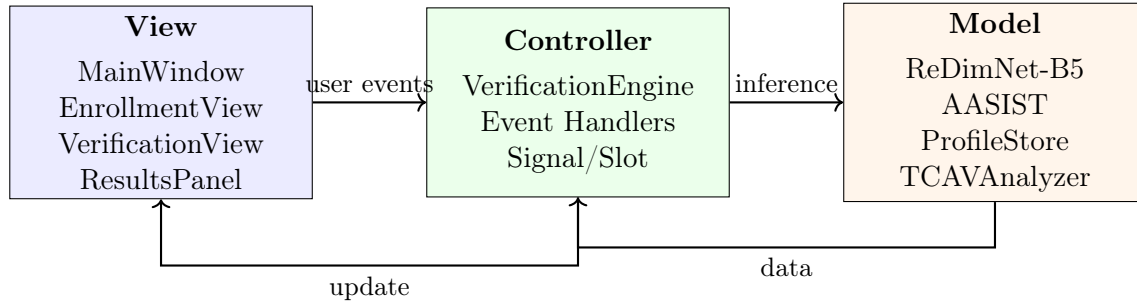


Figure 14: Model-View-Controller architecture

4.5 Testing Platform

The testing strategy encompasses multiple levels to ensure system correctness, robustness, and performance:

Integration Tests: Verify that components work correctly together. Complete enrollment and verification pipelines are tested end-to-end.

Model Validation: Verify ML models meet accuracy specifications. ReDimNet-B5 is validated against VoxCeleb (target EER < 1%). The deepfake detector is validated against synthetic samples generated with F5-TTS.

Robustness Tests: Verify system behavior under adverse conditions including noisy audio, varying durations, and edge cases.

Performance Tests: Verify end-to-end latency remains below 2 seconds.

Table 7 summarizes the testing requirements.

Table 7: Testing requirements and acceptance criteria

Type	Target	Metric	Threshold
Integration	Full pipeline	Pass rate	100%
Model Validation	ReDimNet-B5	EER	<1%
Deepfake Validation	Detector	Accuracy	>95%
Performance	End-to-end	Latency	<2s
UI	All views	Workflows	Pass

5. User Interface

This section presents the user interface screens of the BioVoice application.

5.1 Enrollment Screen

Figure 15 shows the enrollment interface where users create their voice profile. The screen displays the user ID input field with availability validation, real-time waveform visualization during recording, and a recording timer.

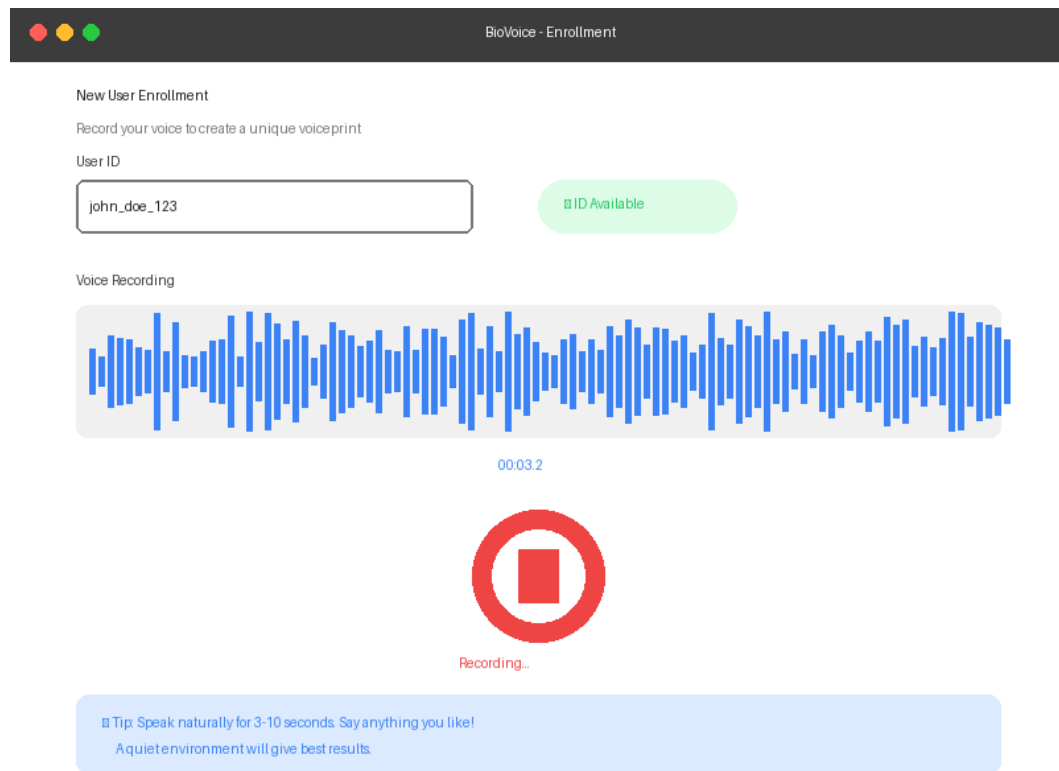


Figure 15: Enrollment screen with voice recording interface

5.2 Processing Screen

Figure 16 shows the audio processing screen displayed while the system converts the voice recording into features. Progress indicators show the current preprocessing stage completion.

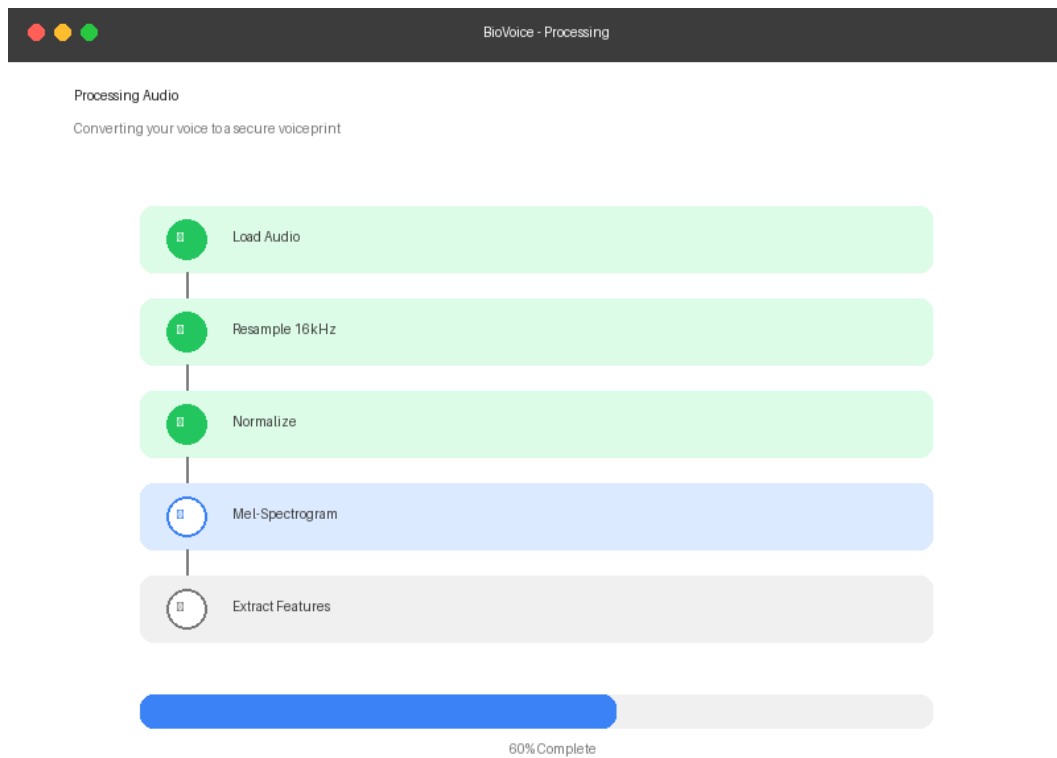


Figure 16: Audio processing screen showing pipeline progress

5.3 Deepfake Detection Result Screen

Figure 17 shows the deepfake detection result, displaying the authenticity analysis with detailed scores for voice naturalness, spectral consistency, temporal patterns, and artifact detection.

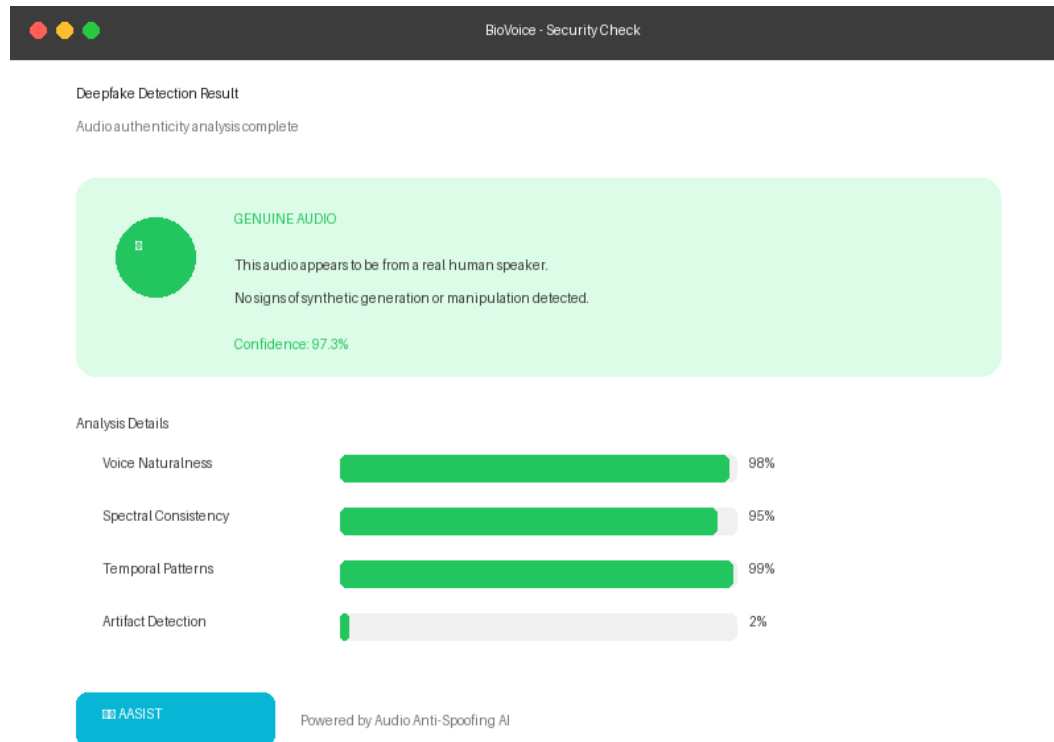


Figure 17: Deepfake detection result showing genuine audio analysis

5.4 Verification Result Screen

Figure 18 shows the verification result after successful identity verification. The screen displays the similarity score gauge, authenticity check status, and action buttons for continuing or retrying.

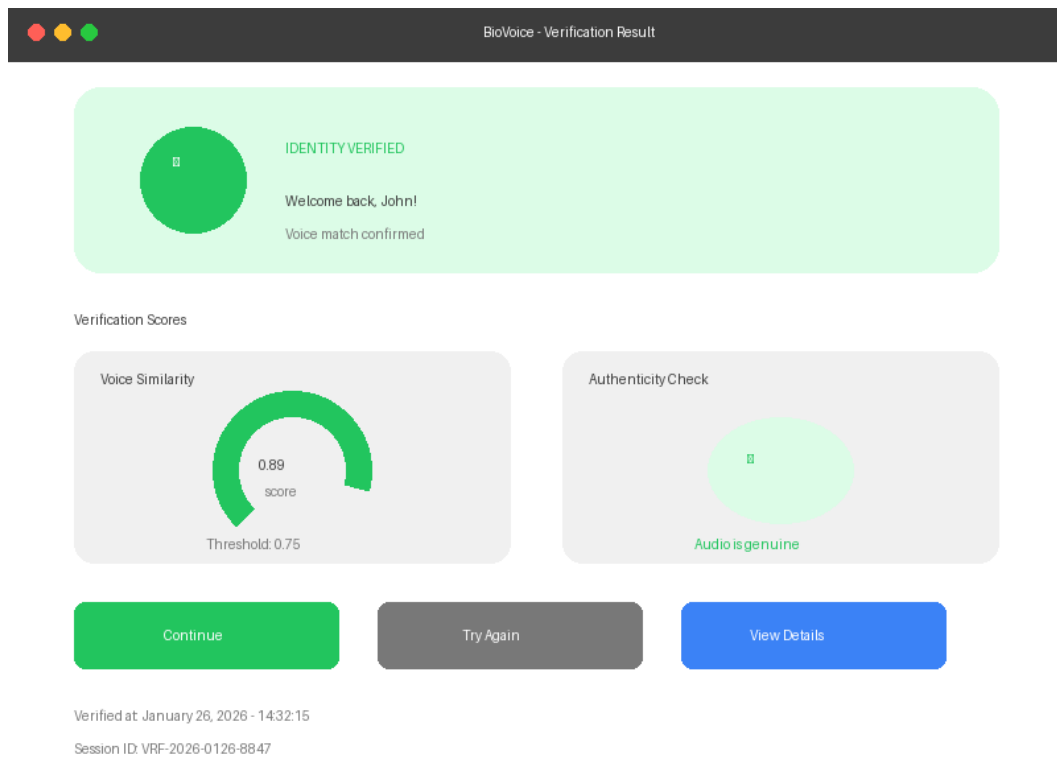


Figure 18: Verification result screen showing successful authentication

5.5 TCAV Explanation Screen

Figure 19 shows the TCAV explanation interface providing interpretable insights into the verification decision. The screen displays contributing voice concepts with their relative weights.

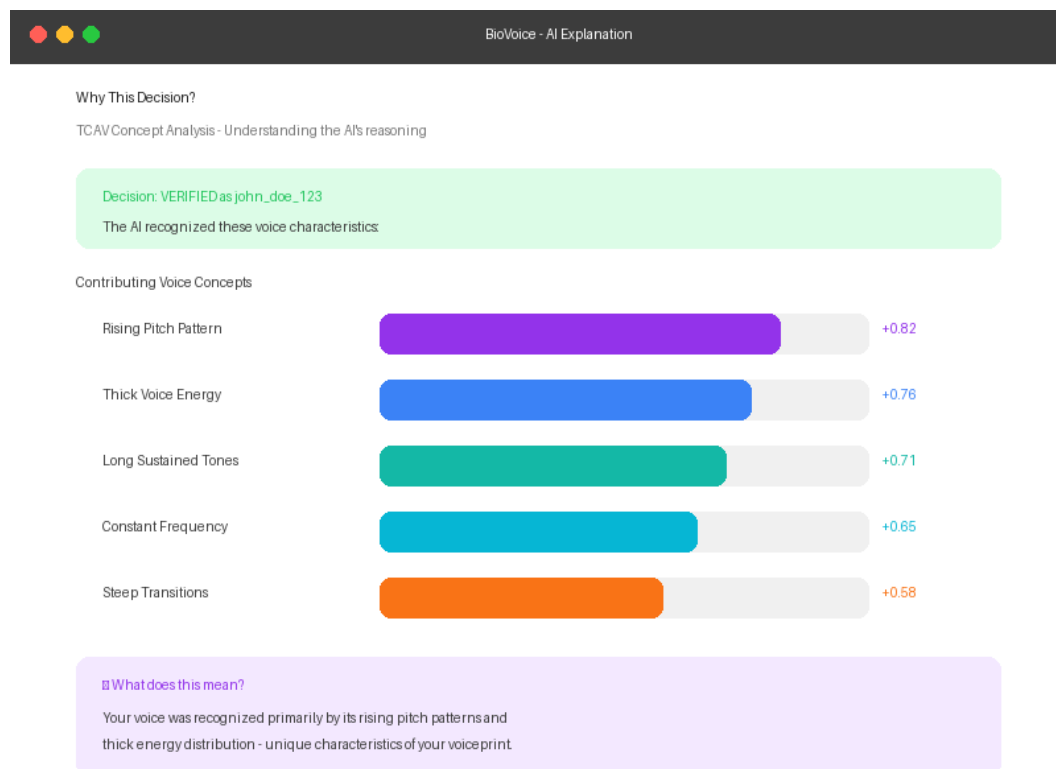


Figure 19: TCAV explanation screen showing voice concept contributions

5.6 Deepfake Generator (Testing)

Figure 20 shows the deepfake generator interface used for testing. This screen allows generation of synthetic audio samples to validate AASIST detection capability.

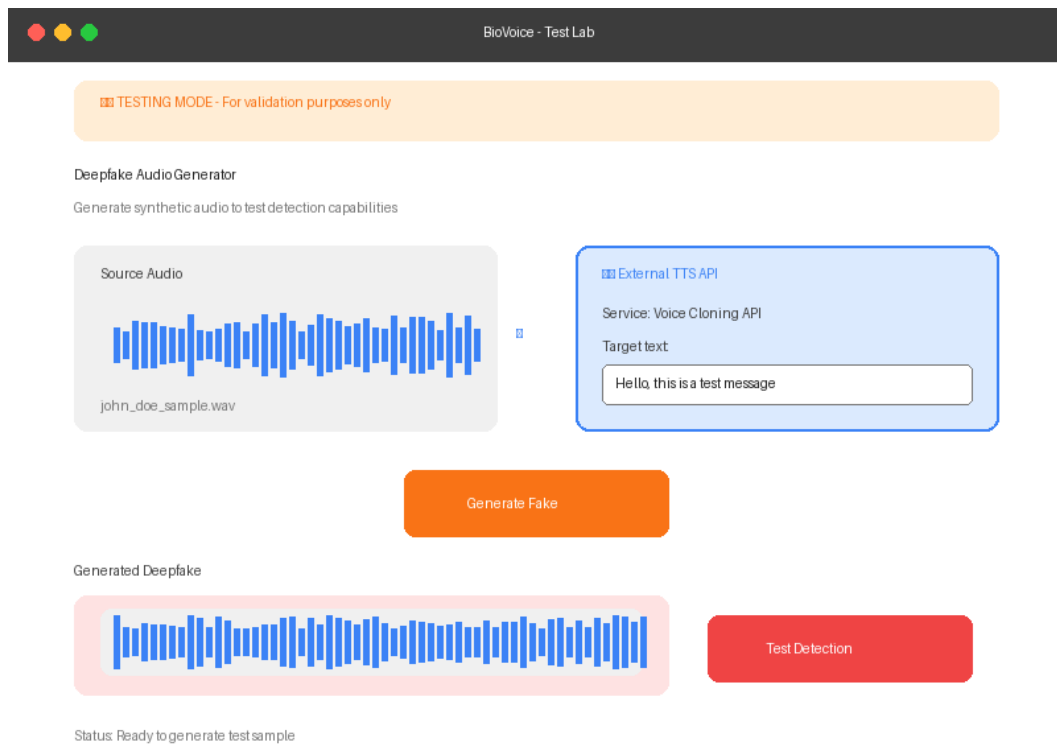


Figure 20: Deepfake generator screen for testing and validation

6. Appendix-A: TCAV Explainability

6.1 Acoustic Concept Taxonomy

14 concepts defined by 4 attributes:

Table 8: Acoustic concept attributes

Duration	Contour	Slope	Thickness
short, long	constant, rising, dropping	flat, steep	thin, thick

Example: `long_rising_steep_thick` = sustained ascending pitch with steep transitions and broad energy distribution.

6.2 Layer-wise Analysis Results

Table 9 shows the layer-wise analysis results. STAGE4 was selected based on highest ARI and silhouette scores.

Table 9: Layer-wise TCAV analysis results (STAGE4 optimal)

Layer	ARI	Silhouette	PC1%	PC2%	Total
STEM	-0.011	0.232	51.0	9.3	60.3%
STAGE0	0.133	-0.120	75.0	3.5	78.4%
STAGE1	0.065	0.219	85.5	2.9	88.4%
STAGE2	0.117	0.209	71.1	4.2	75.3%
STAGE3	0.095	0.294	85.4	5.3	90.7%
STAGE4	0.480	0.400	96.1	1.8	97.9%
STAGE5	0.106	0.391	89.0	3.1	92.1%

6.3 TCAV Integration

TCAV explanations are available on-demand via the `explain()` method:

- **Standard verification:** Returns decision only (50ms)
- **With explanation:** Returns decision + TCAV concept scores (120ms)

7. Appendix-B: Team

The BioVoice project is developed by a team of three members who collaborate across all aspects of the system.

Table 10: Project team members

Member	Role
Eden Adiv	ML Engineer
Idan Shavit	ML Engineer
Yoav Zucker	ML Engineer

8. Appendix-C: Schedule

Table 11 presents the project schedule.

Table 11: Project schedule

Phase	Task	Duration	Status
1	Research & Literature Review	10 weeks	Complete
2	Core Development	10 weeks	Complete
3	POC Implementation	6 weeks	Complete
4	Deepfake Research & Testing	5 weeks	Active
5	Validation & Benchmarking	3 weeks	Planned
6	Alpha	4 weeks	Planned
7	Beta	6 weeks	Planned
8	Improvement	4 weeks	Planned
9	Documentation	4 weeks	Planned
Total		52 weeks	

References

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